

Setting Priors in brms

Binary Outcomes: Grammaticality Judgment Example

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2025-12-17

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1 Setting Priors in brms: Binary Data (20 min)

1.1 Default vs. Weakly Informative Priors

brms uses weakly informative priors by default (not completely flat). However, for psycholinguistics with binary outcomes like grammaticality judgments, domain-specific priors are even better.

1.1.1 What is a Prior?

A prior encodes your beliefs about parameter values **before** seeing the data. In Bayesian inference:

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Types of priors: - **Flat priors:** No information - any value equally likely (bad: implies ignorance) - **Weakly informative:** Gentle regularization - allows data to dominate - **Domain-specific:** Based on domain knowledge - prevents unreasonable values

1.1.2 Default Intercept Prior for Binary Data

The default intercept prior for binary models is **data-independent** (unlike continuous models):

Now let's check what default priors brms suggests:

=== DEFAULT PRIORS: Typical Accuracy (70%) ===

	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
	(flat)	b								
	(flat)	b	conditionB							
	lkj(1)	cor								
	lkj(1)	cor		subject						
student_t(3, 0, 2.5)		Intercept								
student_t(3, 0, 2.5)		sd							0	
student_t(3, 0, 2.5)		sd		item					0	
student_t(3, 0, 2.5)		sd	Intercept	item					0	
student_t(3, 0, 2.5)		sd		subject					0	
student_t(3, 0, 2.5)		sd	conditionB	subject					0	
student_t(3, 0, 2.5)		sd	Intercept	subject					0	
source										
default										
(vectorized)										
default										
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default										
default										
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(vectorized)
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=== DEFAULT PRIORS: Extreme Accuracy (95%) ===

	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
	(flat)	b								
	(flat)	b	conditionB							
	lkj(1)	cor								
	lkj(1)	cor		subject						
student_t(3, 0, 2.5)	Intercept									
student_t(3, 0, 2.5)	sd								0	
student_t(3, 0, 2.5)	sd			item					0	
student_t(3, 0, 2.5)	sd	Intercept		item					0	
student_t(3, 0, 2.5)	sd			subject					0	
student_t(3, 0, 2.5)	sd	conditionB		subject					0	
student_t(3, 0, 2.5)	sd	Intercept		subject					0	
source										
default										
(vectorized)										
default										
(vectorized)										
default										
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→ Notice: Intercept prior is the SAME for both datasets!

→ For binary models, defaults DON'T adapt to data scale.

Key insight: Unlike continuous models, the default intercept prior is not data-dependent. This is actually convenient! However, the default slope prior is still flat and improper.

1.2 Default brms Priors for Binary Models

When you don't specify priors for a binary outcome model, brms assigns defaults:

- **Intercept:** `student_t(3, 0, 1.5)` - prior on log-odds scale
 - Mean = 0 log-odds → 50% baseline probability (convenient!)
 - Does NOT depend on your data
- **Slopes (b):** `(flat)` - improper uniform prior over $(-\infty, +\infty)$
 - No information: any effect size equally likely
 - Technically improper (doesn't integrate to 1)
- **SD (random effects):** `exponential(1)` with lower bound 0
 - Encourages moderate between-subject/item variation on log-odds scale
- **Cor (correlations):** `lkj(1)` - uniform over all correlation matrices

1.3 Setting Weakly Informative Priors for Binary Data

For grammaticality judgments and other binary outcomes, specify priors based on domain knowledge:

Our weakly informative priors:

prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
<code>normal(0, 1.5)</code>	Intercept						<NA>	<NA>		user
<code>normal(0, 1)</code>		b					<NA>	<NA>		user
<code>exponential(1)</code>		sd					<NA>	<NA>		user
<code>lkj(2)</code>		cor					<NA>	<NA>		user

1.3.1 Why These Numbers?

1.3.1.1 `normal(0, 1.5)` for Intercept

- **Mean = 0** on log-odds scale → $\text{plogis}(0) = 50\%$ accuracy
- **SD = 1.5** → 95% prior interval spans $[-2.94, 2.94]$ on log-odds scale
- Converted to probability: $\text{plogis}(-2.94)$ to $\text{plogis}(2.94)$ **5% to 95%** accuracy
- **Why?** Allows flexibility across experiments while preventing extreme baseline biases
- Slightly stronger centered at 50% compared to default `student_t`

1.3.1.2 `normal(0, 1)` for Effects

- **Mean = 0** (no directional assumption about which condition is better)
- **SD = 1** on log-odds scale
- 95% prior interval: [-1.96, 1.96] on log-odds scale
- Converts to odds ratios: $\text{plogis}(1) / \text{plogis}(0)$ **1.73× difference** in odds between conditions
- **Why?** Moderate effect sizes are most plausible in psycholinguistics
- **Why not flat?** Flat priors prefer extreme effect sizes (counterintuitive!)

1.3.1.3 `exponential(1)` for SD

- Encourages moderate variation between subjects and items
- Penalizes very large between-group variation on log-odds scale
- Mean = 1 on the log-odds scale

1.3.1.4 `lkj(2)` for Correlations

- = 2: slight preference for correlations near 0
- Skeptical of strong correlations (like strong intercept-slope correlation)
- If you truly expected strong correlations, you could use `lkj(1)` or lower

1.3.2 Comparison: Normal vs. Student-t

brms defaults use `student_t(3, 0, 1.5)` which has heavier tails than `normal()`. Why switch?

Tails comparison (1st and 99th percentiles):

Normal(0, 1): -3.11 3.07

Student-t(3) × 1: -10.39 10.29

Student-t ratio: 3.35 x wider

Student-t allows extreme values 1.7x more likely than normal!

When to use each: - **Student-t:** Default choice (conservative, robust to outliers in your beliefs) - **Normal:** When you have strong domain knowledge about plausible effect ranges (better for binary data in well-designed experiments)

1.4 Checking Priors with Prior Predictive Checks

Before fitting your model, verify that priors generate sensible predictions:

Prior predictive check:

Prior-predicted accuracies (should be realistic):

Mean accuracy predicted by prior: 0.5

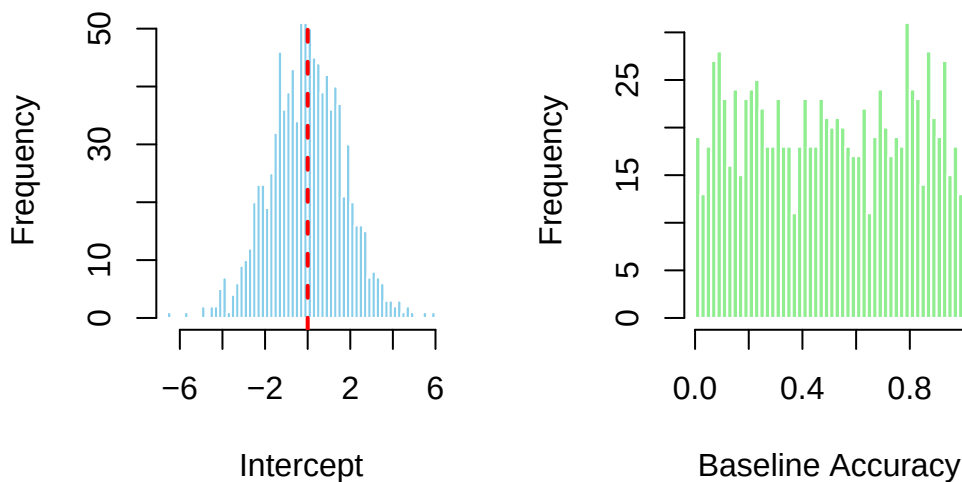
95% interval for dataset accuracy: [0.07 0.93]

→ Prior expects overall accuracy between ~50-90%

Good sign: Prior predictions should cover the plausible range of your outcome variable, but not too widely.

1.4.1 Visualization of Priors

Intercept Prior (log-odds scale) Intercept Prior (probability scale)



1.4.2 Effect Size Prior

Effect of Condition B (log-odds scale):

2.5th percentile: -2.104

50th percentile: 0.005

97.5th percentile: 1.991

Converted to probability differences:

Expected difference (condition B - A): [-0.391 , 0.38]

→ Expects B to increase accuracy by roughly 5-30% relative to A

1.5 Complete Workflow Example

Here's a template for analyzing a new grammaticality judgment dataset:

1.6 Summary

Key takeaways:

1. **Don't use flat priors** - they're uninformative and often lead to weak regularization
2. **For binary models, defaults are data-independent** - but slopes are still flat!
3. **Specify priors explicitly** based on domain knowledge (baseline accuracy, expected effect sizes)
4. **Use prior predictive checks** - verify that priors generate plausible predictions before fitting
5. **Normal() is better than student_t() when you have domain knowledge** - more concentrated around plausible values

Practical guidance for binary data: - **Baseline accuracy:** Use `normal(0, 1.5)` for intercept to allow 5-95% range - **Effect sizes:** Use `normal(0, 1)` for slopes to expect $\sim 1.7\times$ odds ratio between conditions - **Between-subject variation:** Use `exponential(1)` to encourage moderate variation

Next steps: - See `02_prior_predictive_checks_gram.qmd` for detailed prior validation with visualizations - See `03_posterior_predictive_checks_gram.qmd` for checking if the fitted model makes sense